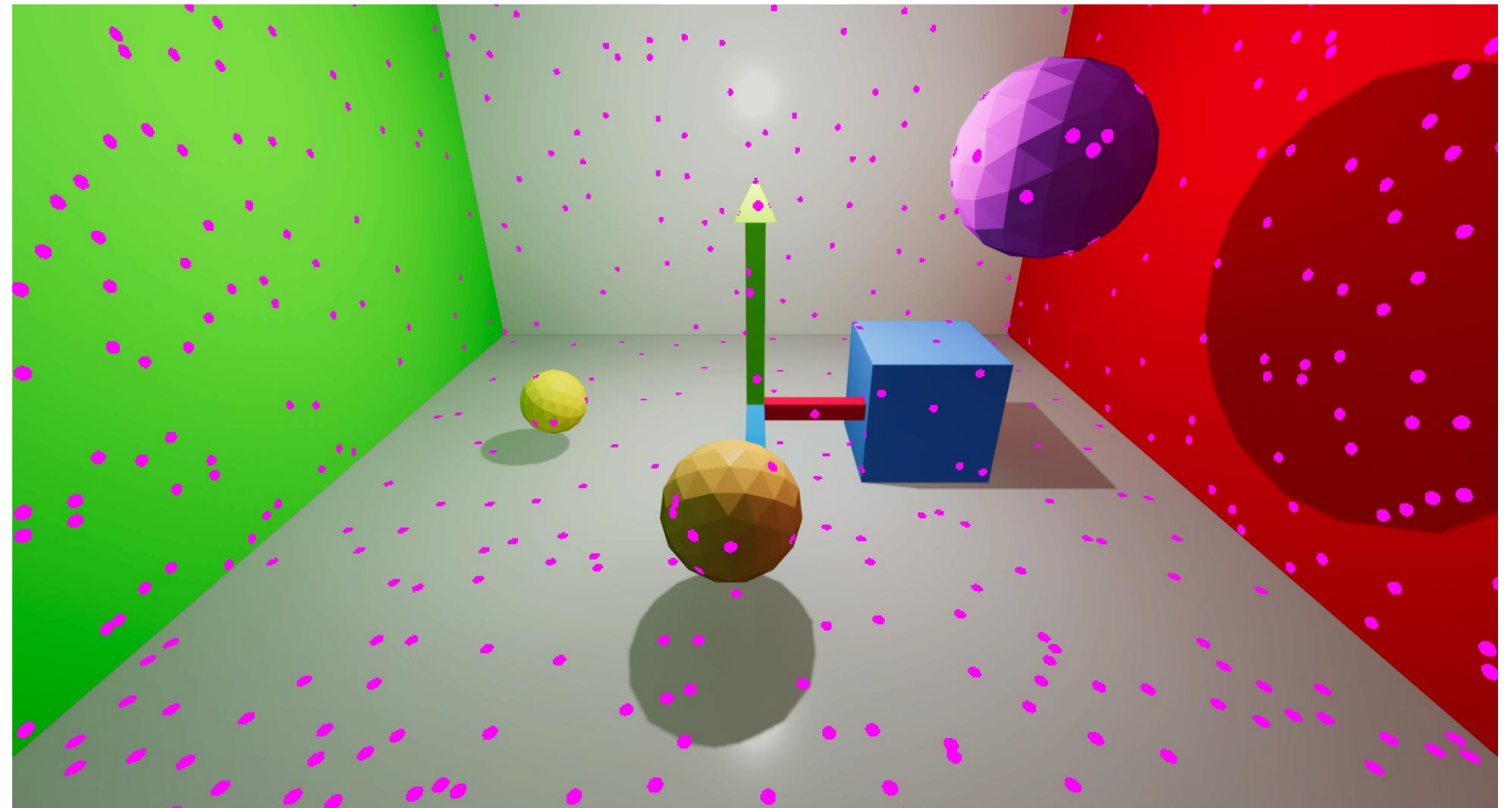
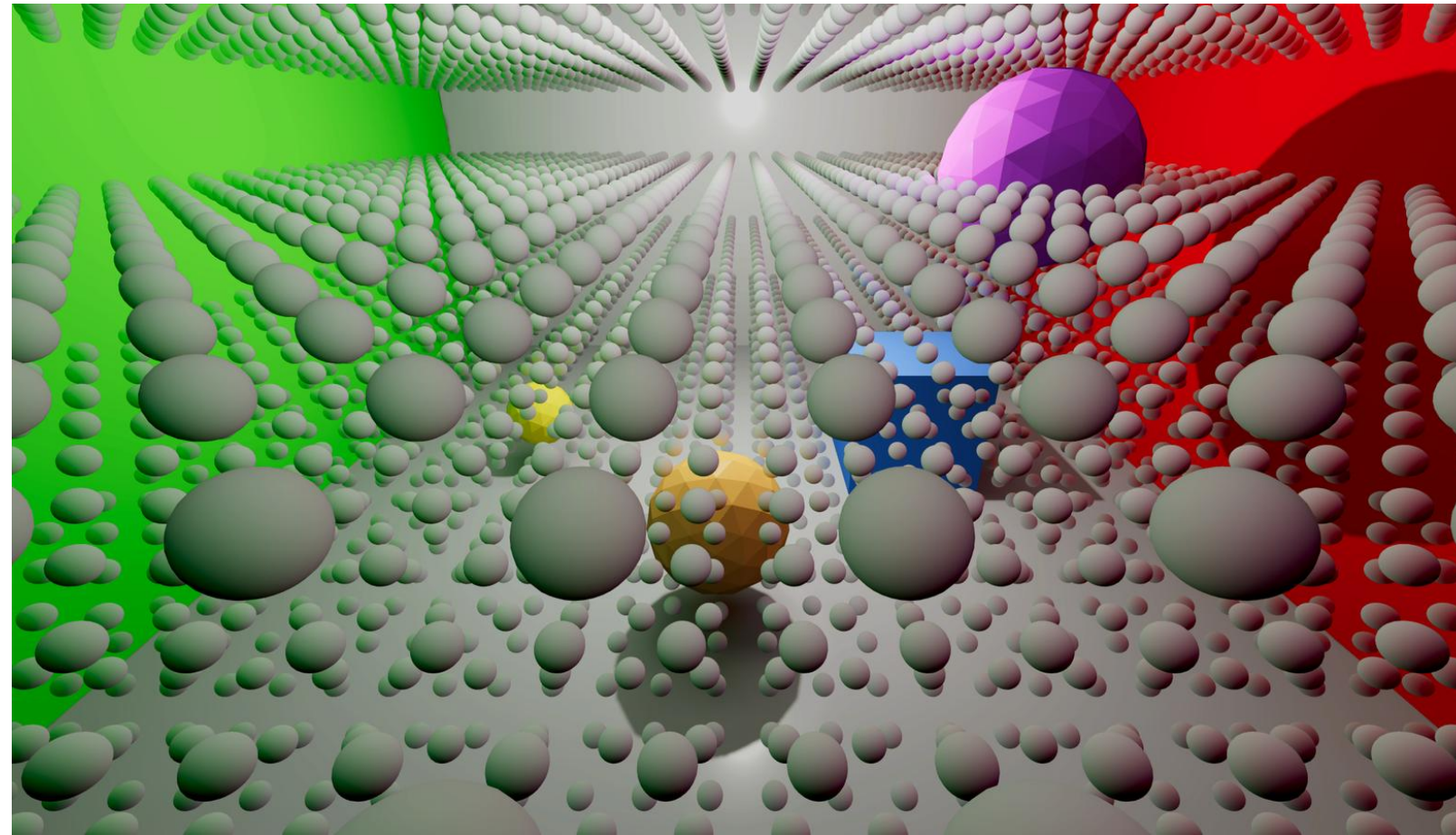


Voxel-based Global Illumination

Voxel-based Global Illumination (Voxel GI)



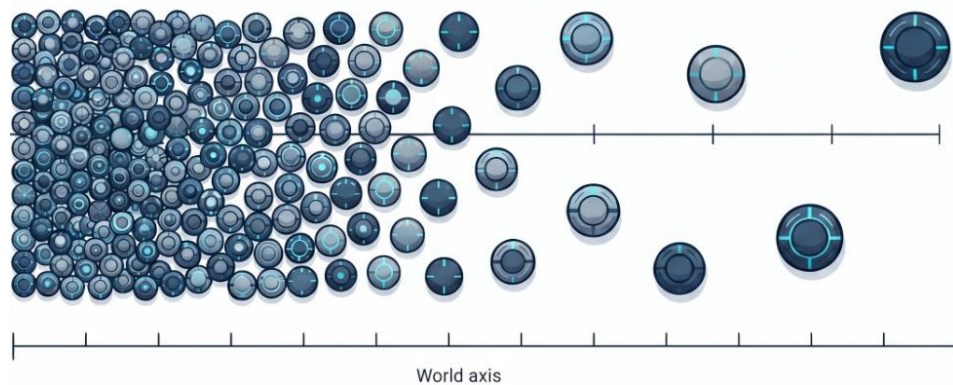
uniform-resolution

Near: High Resolution (High Density)

GI elements are tightly packed to capture intricate lighting and shadow details near the viewer. Adaptive coverage logic spawns more elements in high-frequency areas.

Far: Low Resolution (Low Density)

Elements become increasingly sparse as distance increases to conserve memory and processing power. Multi-level clipmaps represent large spaces with decreasing resolution.



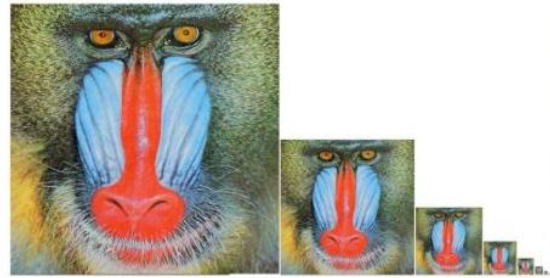
DDGI / Surfel GI limitation:

lower resolution **reduces detail**, while higher resolution **increases cost**

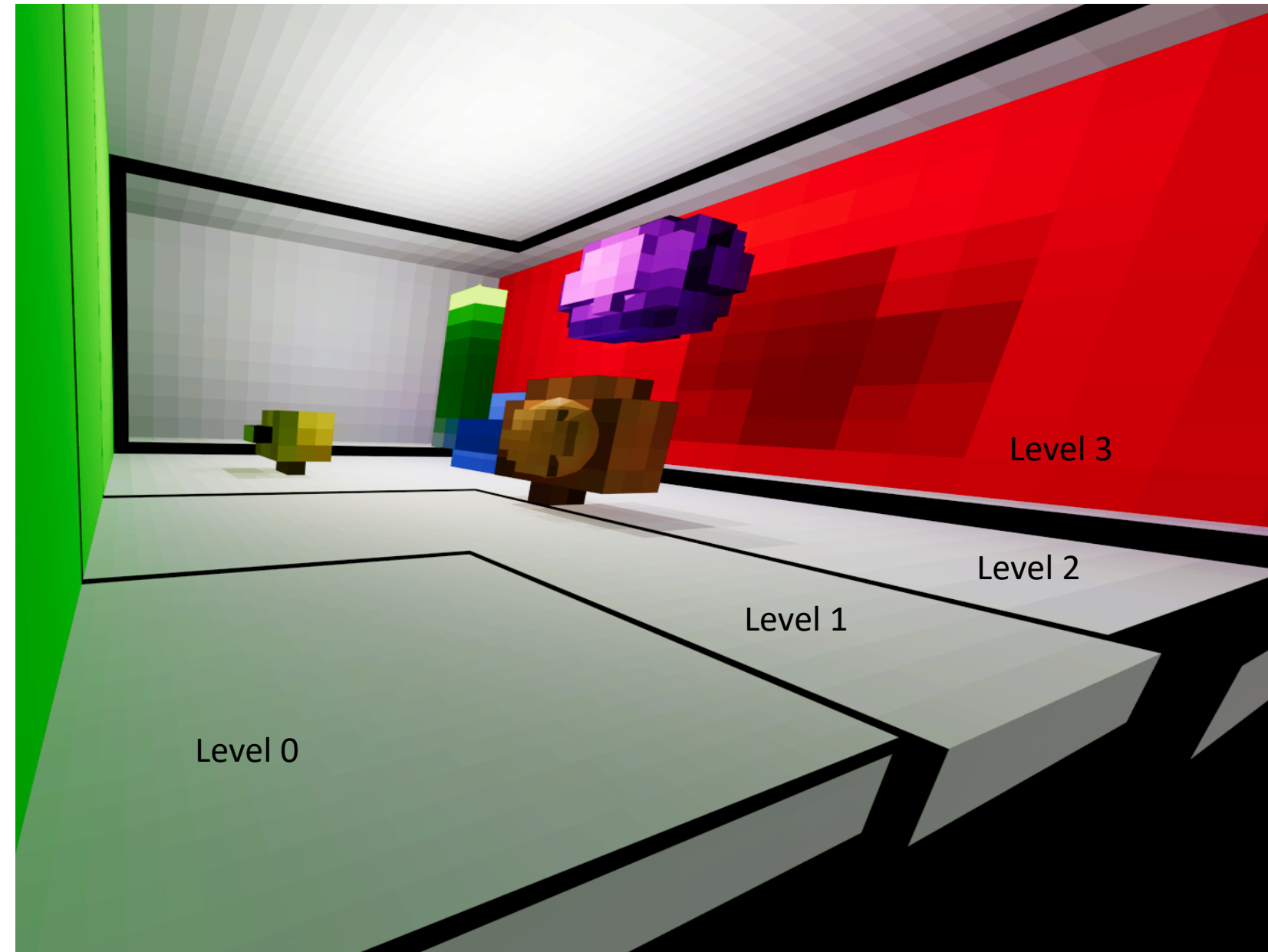
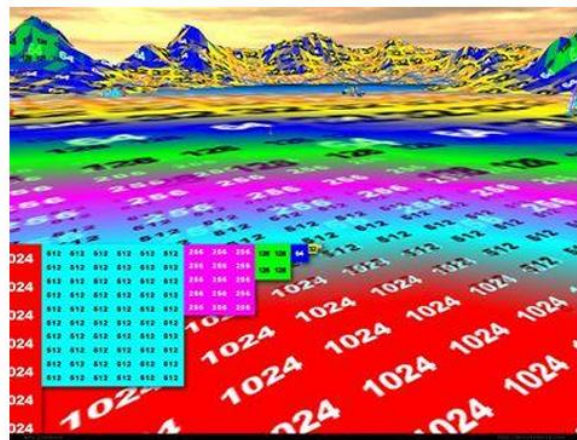
*Resolution = spatial density of probes, surfels, or voxels in world space

Voxel-based Global Illumination (Voxel GI)

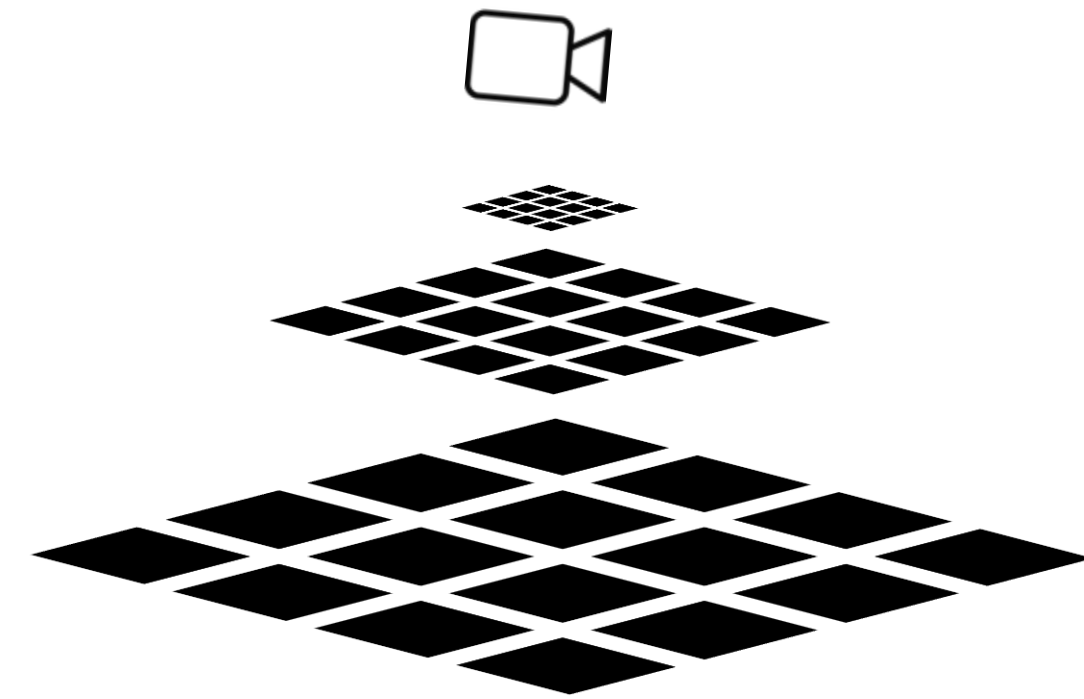
Level of Detail



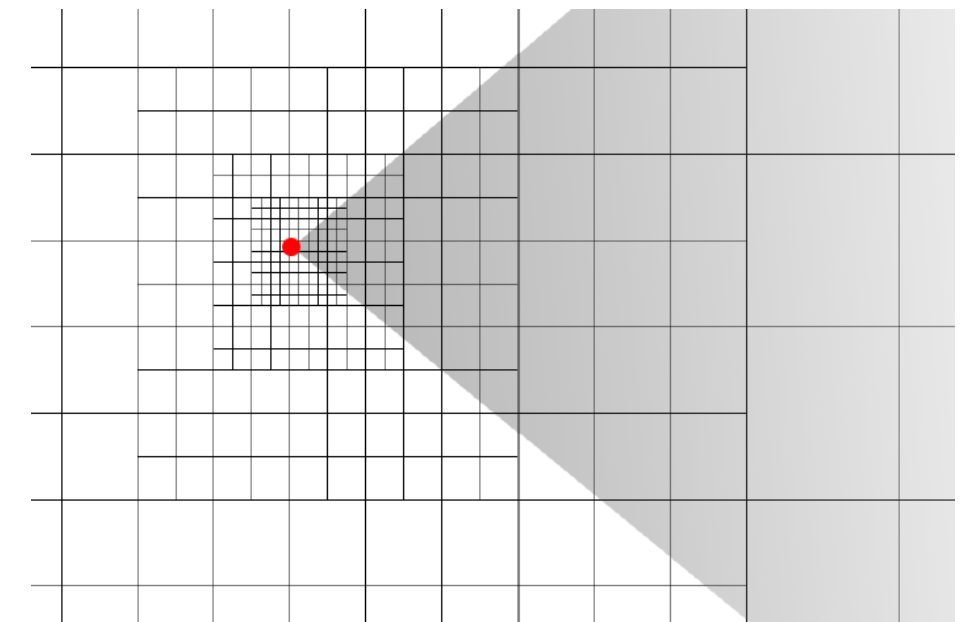
10-level mip map



VoxelGI : multi-resolution



clipmap



Based on the LOD concept, uses 6 distance-based clipmap levels

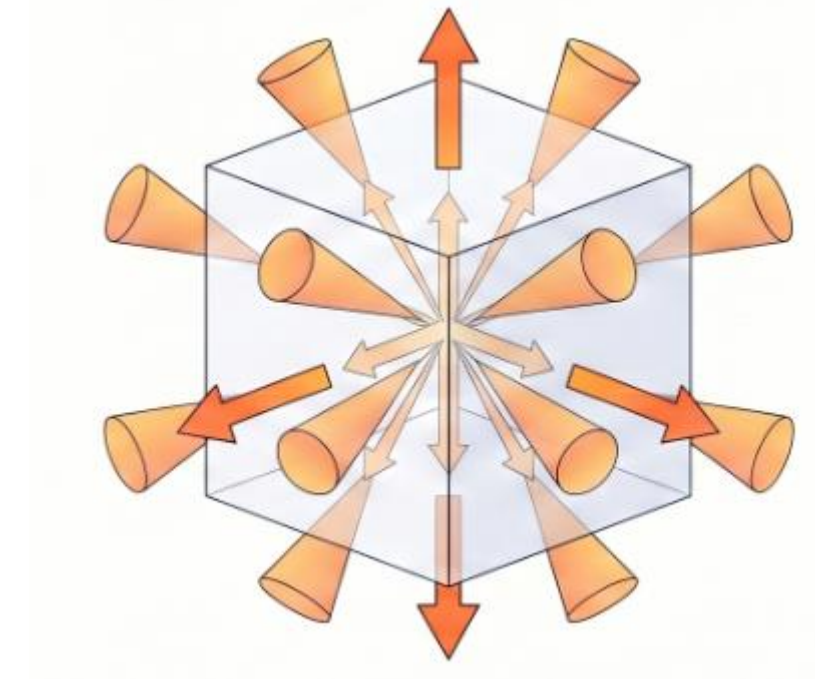


Voxel-based Global Illumination (Voxel GI)

Voxel (Volume + Pixel)

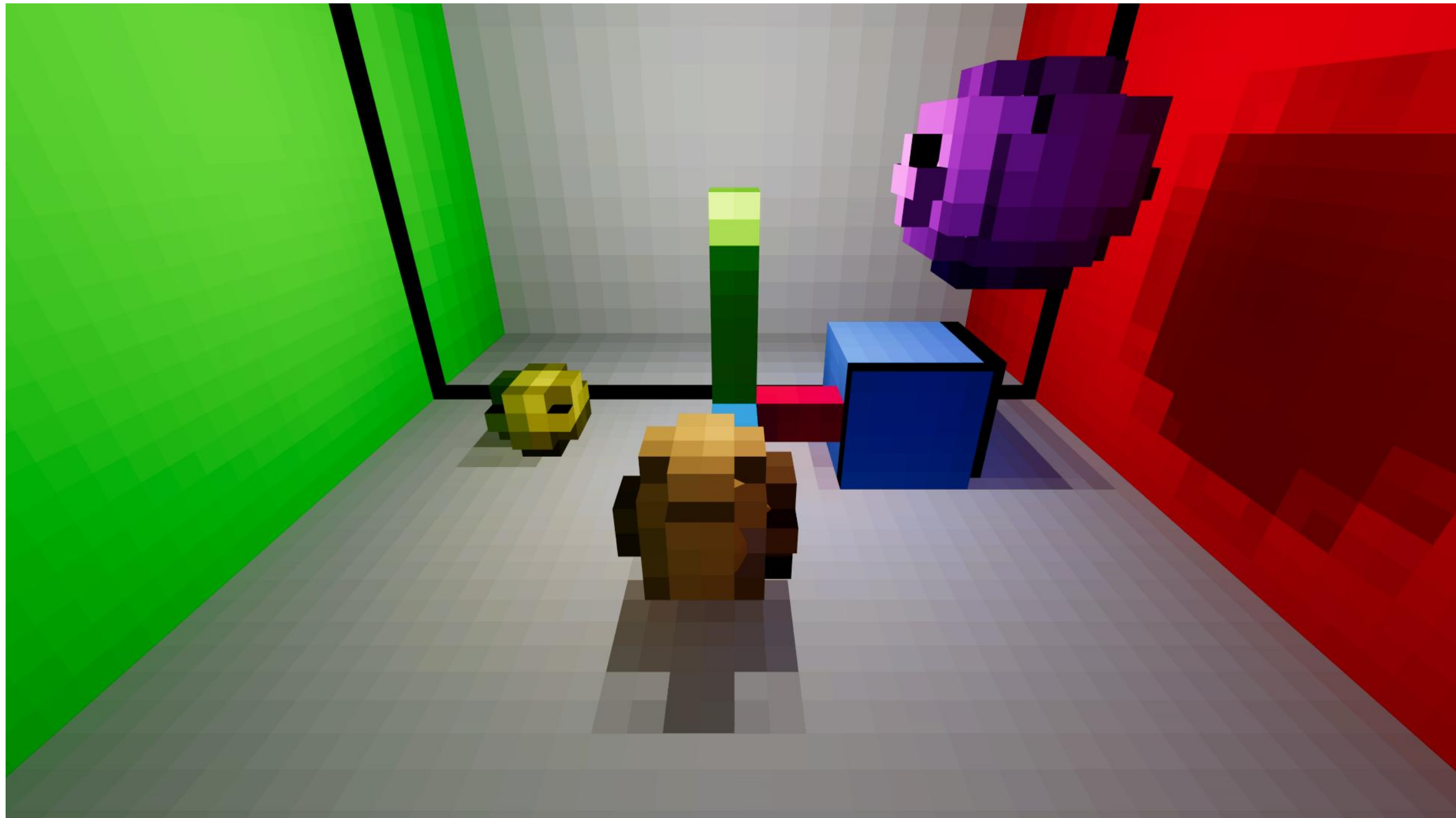


- a unit of a 3D grid-based spatial representation
- Each voxel can store information (e.g., lighting data)



Each voxel stores lighting information (RGBA) separately for 22 directions:
6 axis-aligned directions and 16 cone directions. (not SH L1!)

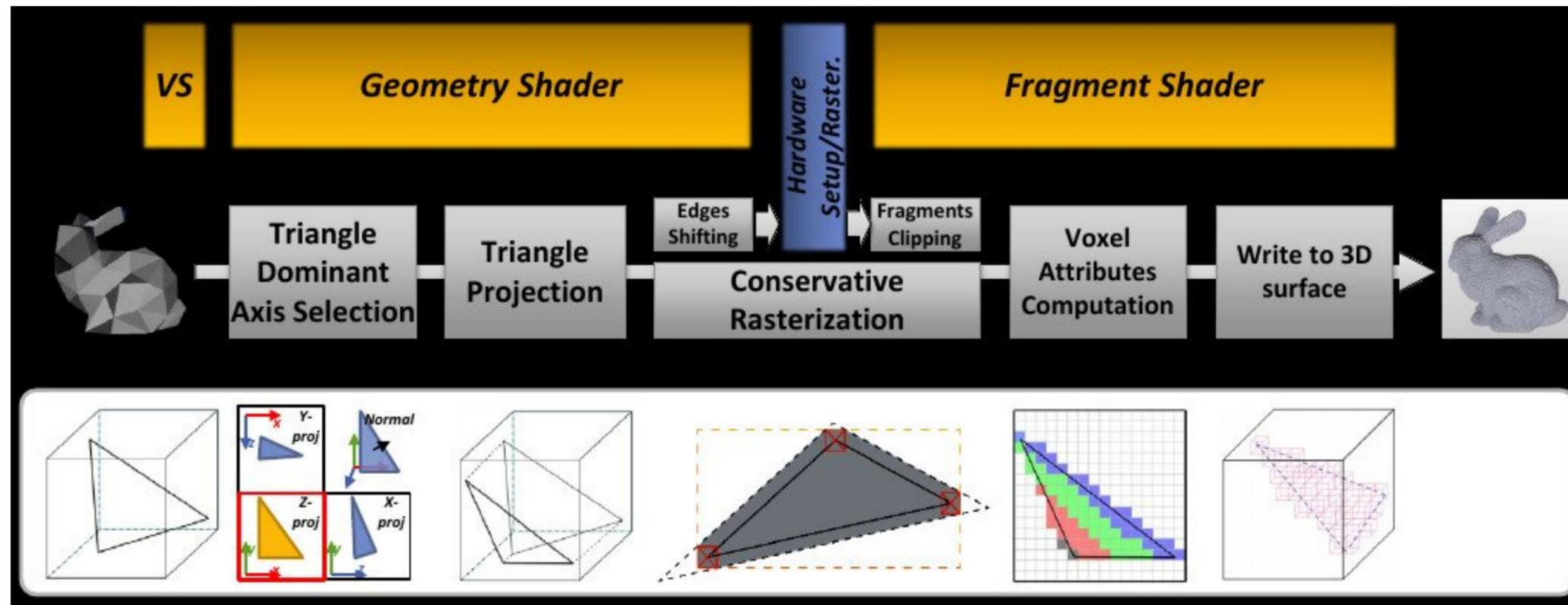
Voxel-based Global Illumination (Voxel GI)



1. **Cone tracing** from each **voxel** → Capture radiance
2. Store **directional radiance** per **voxel** (6 axis directions + 16 cone directions), not SH. (**for specular reflection!**)
3. Reuse the stored radiance during **cone tracing** → Recursive feedback
4. Infinite bounces approximated across frames

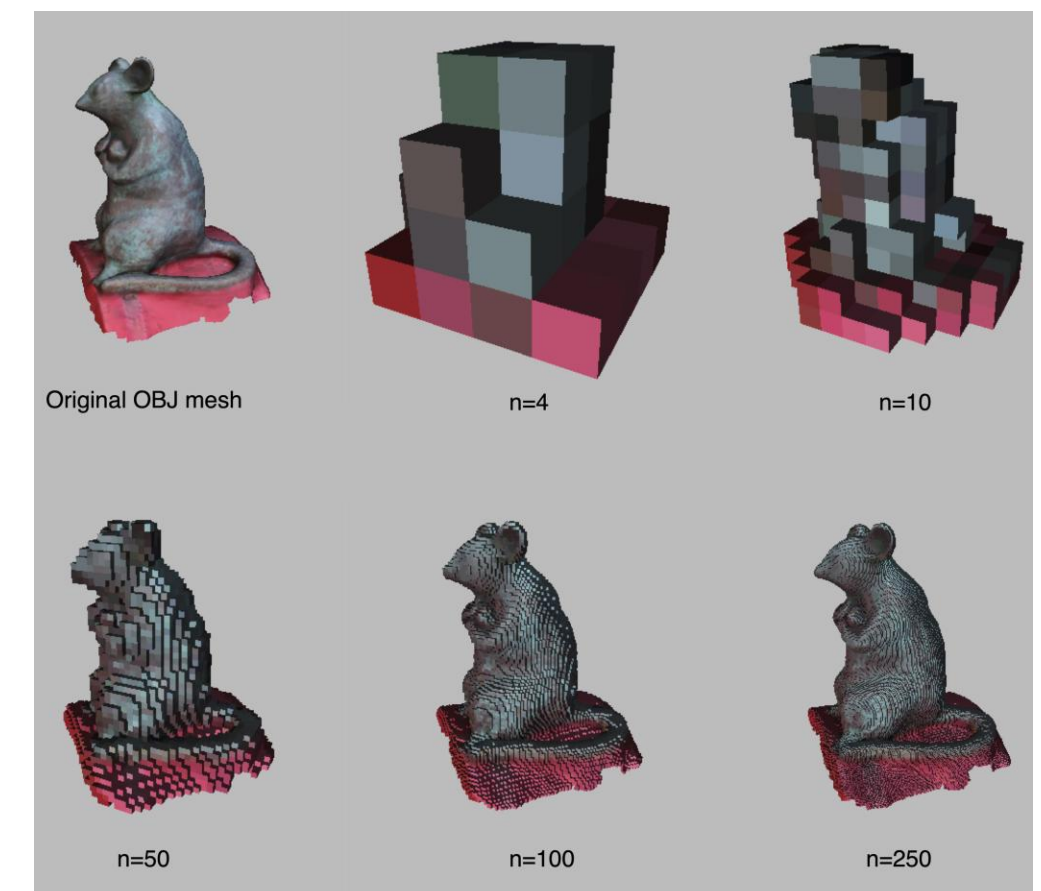
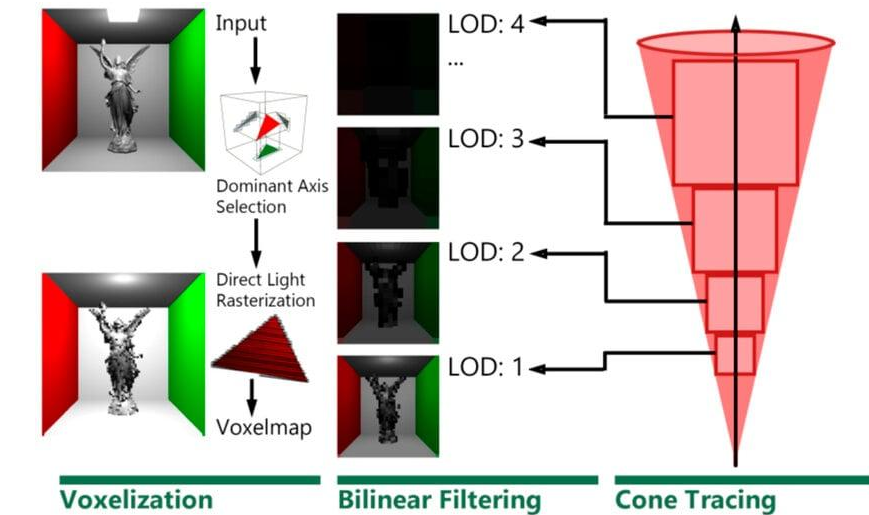
Voxel-based Global Illumination (Voxel GI)

Voxelization



Convert scene geometry into a voxel representation for Voxel GI

- Triangles in view frustum rasterized along their **dominant axis (the most visible)** (X, Y, or Z)
- Rasterization generates occupied voxels + **direct lighting**
- One clipmap level updated per frame
- **Multiple fragments** can contribute to **the same voxel**, accumulating radiance and **opacity** to represent transparency and occlusion.

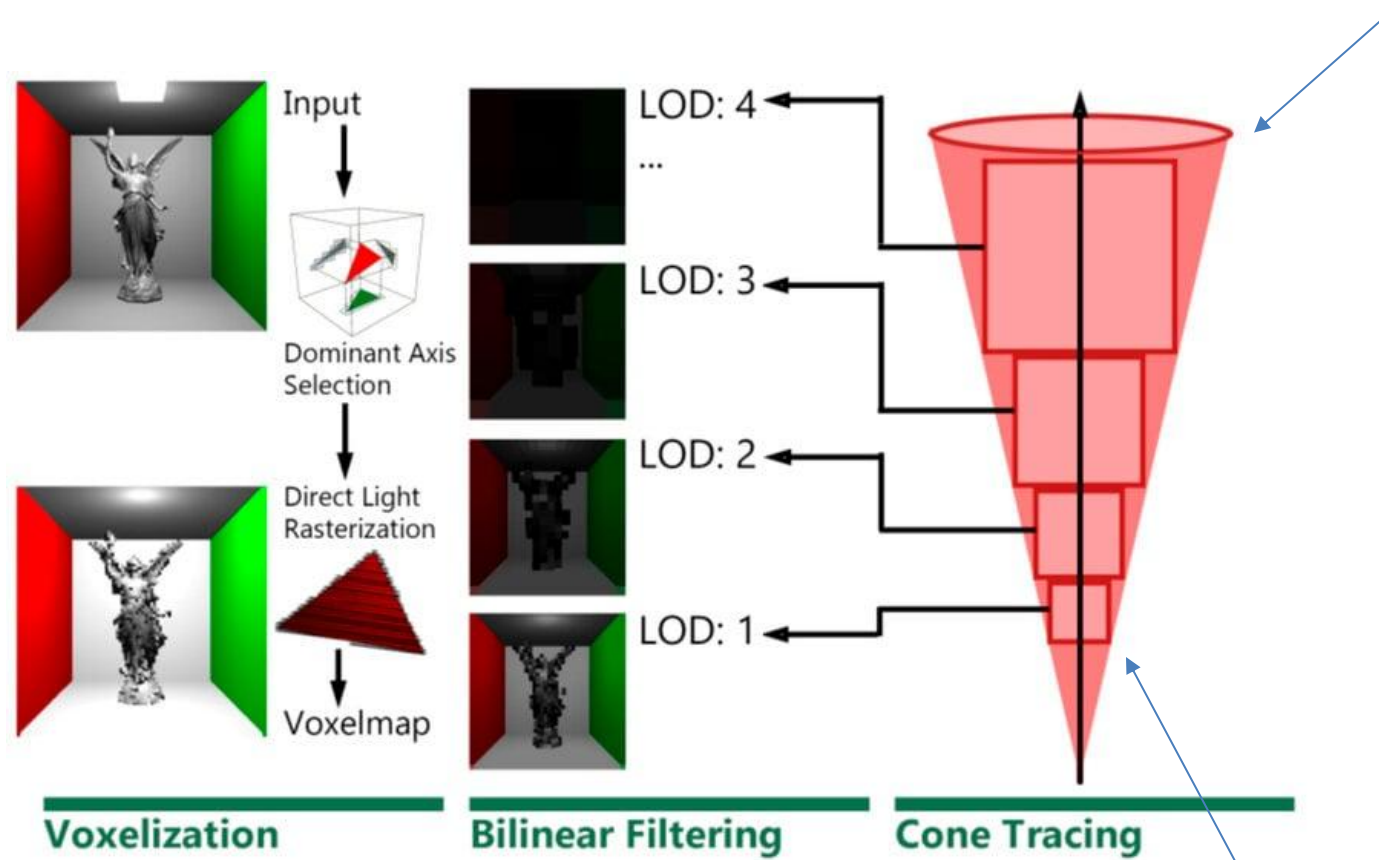


Voxel-based Global Illumination (Voxel GI)

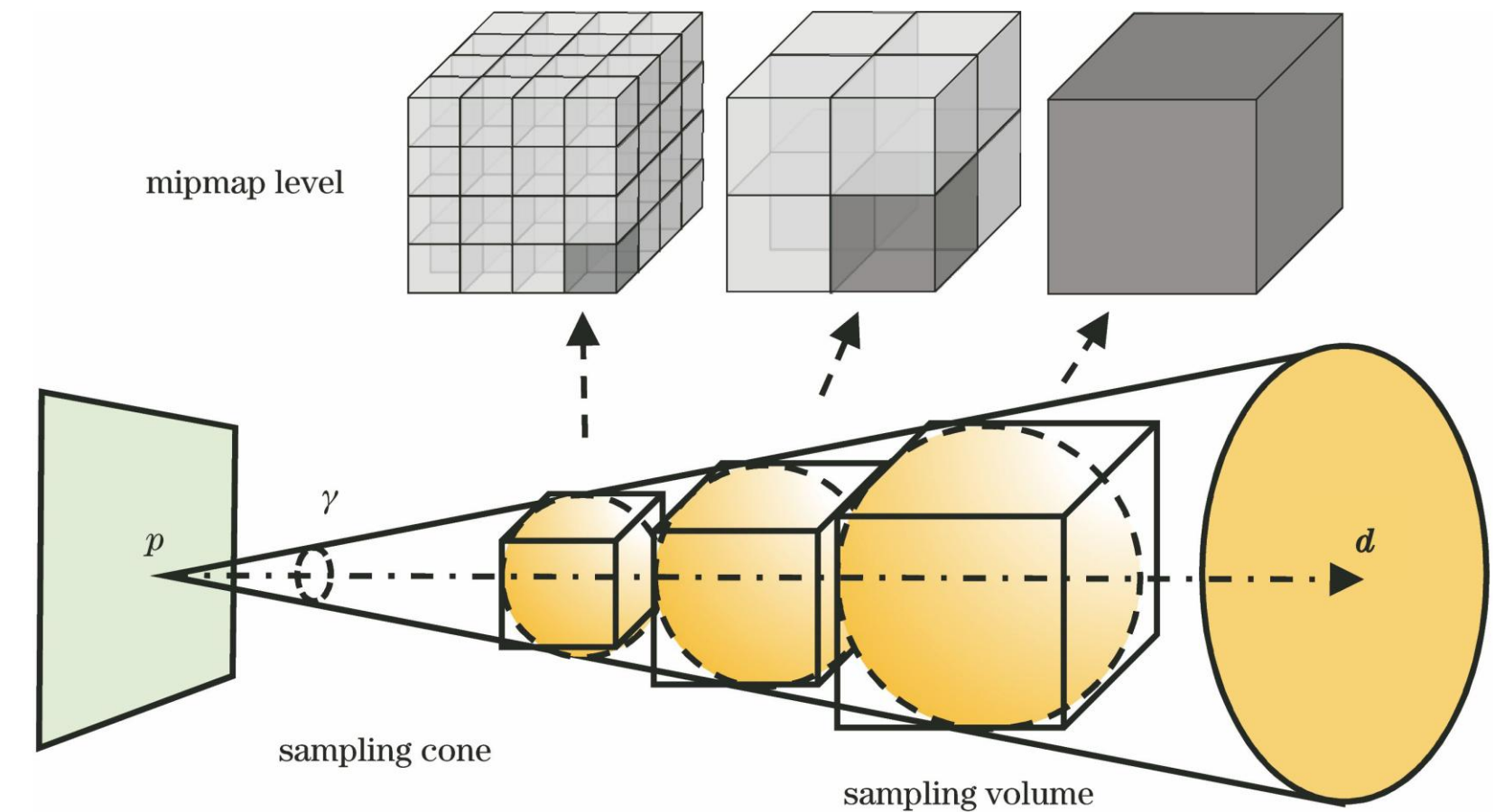
Cone tracing

Voxel GI uses cone tracing instead of ray tracing.

Far regions → higher clipmap levels (**lower-resolution** voxels)



Near regions → lower clipmap levels (**higher-resolution** voxels)



Cone tracing efficiently accumulates directional radiance from voxel clipmaps

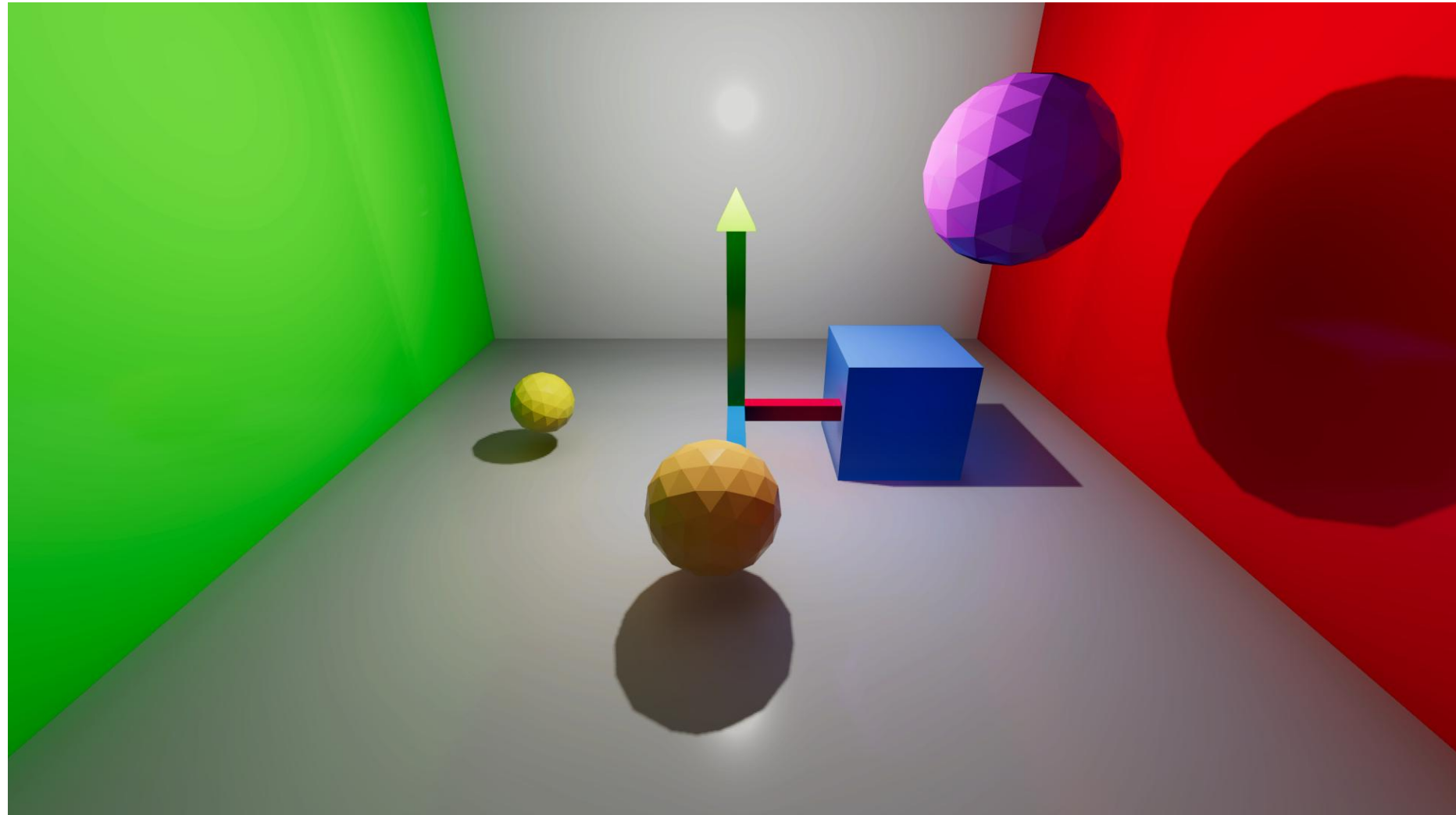
Sampling continues until accumulated **alpha reaches 1.0 (opaque)**

(Each voxel's opacity(alpha) is determined by accumulated fragment coverage during voxelization.)

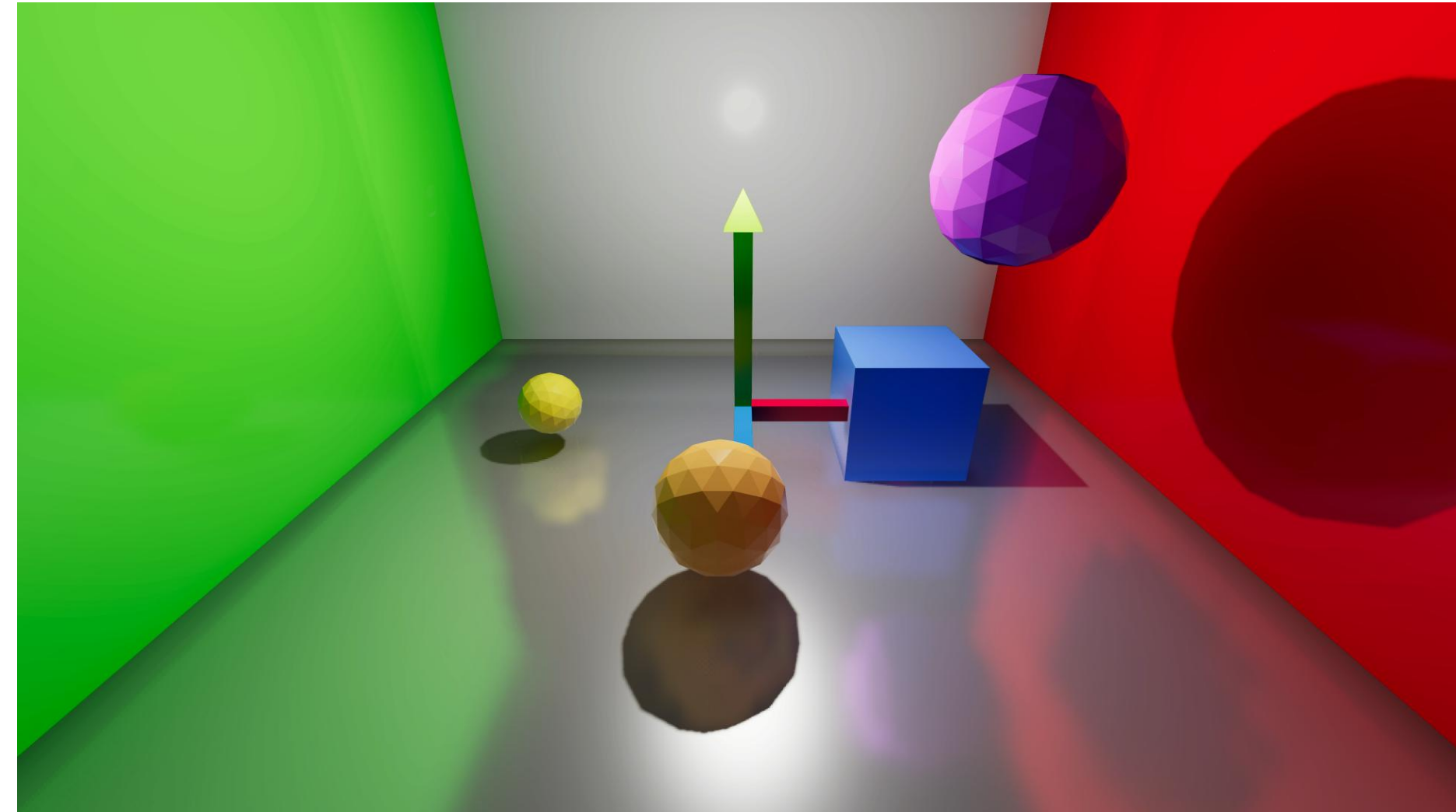
Lighting information integrated through accumulated voxel radiance

Voxel-based Global Illumination (Voxel GI)

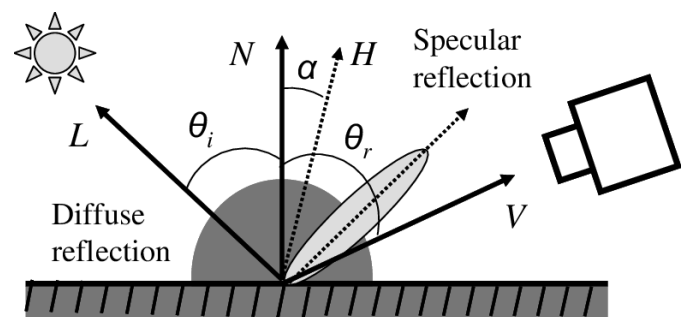
Specular Reflection



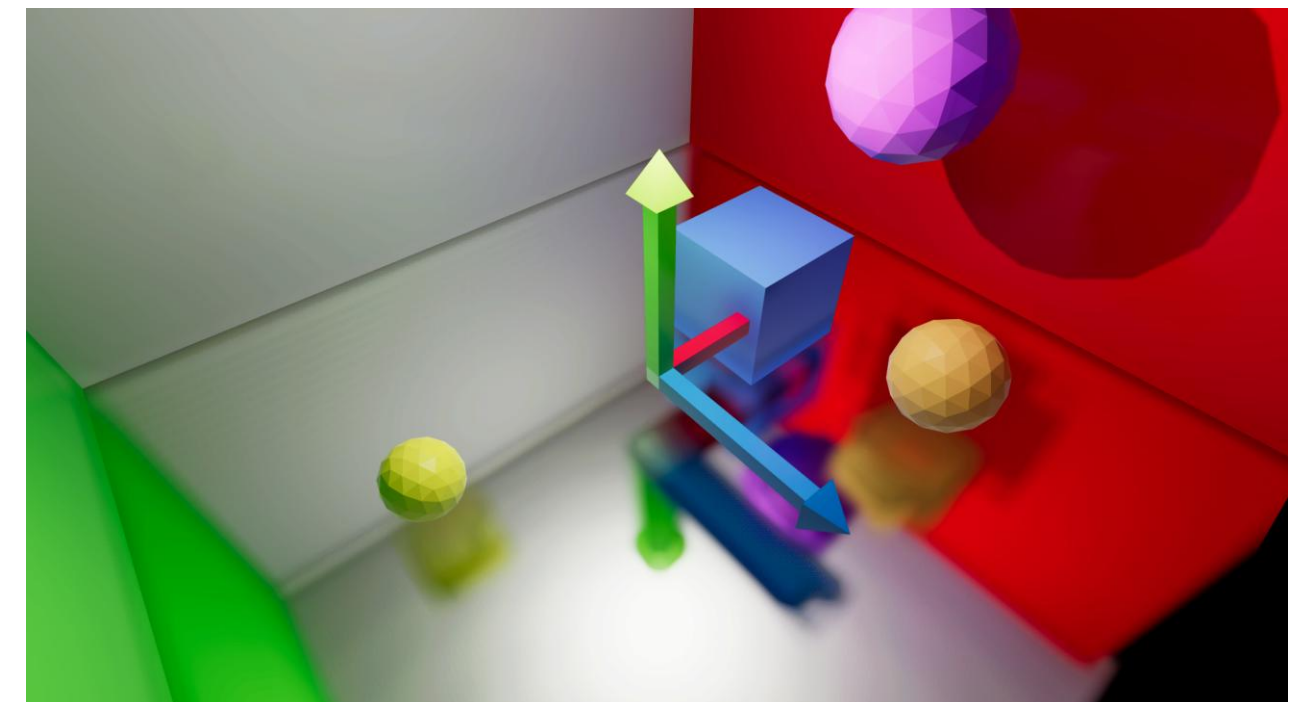
Reflection off



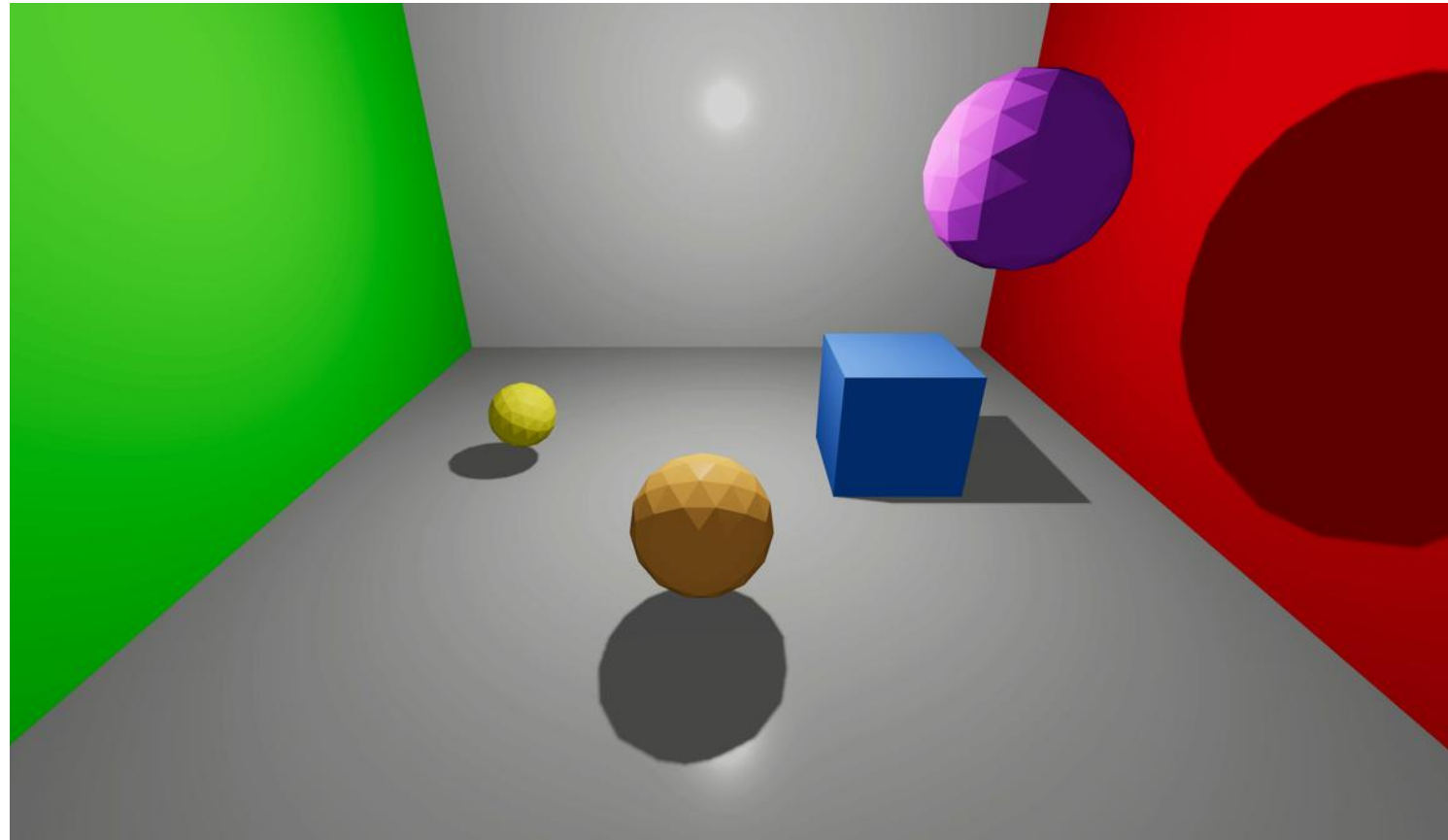
Reflection on (only floor)



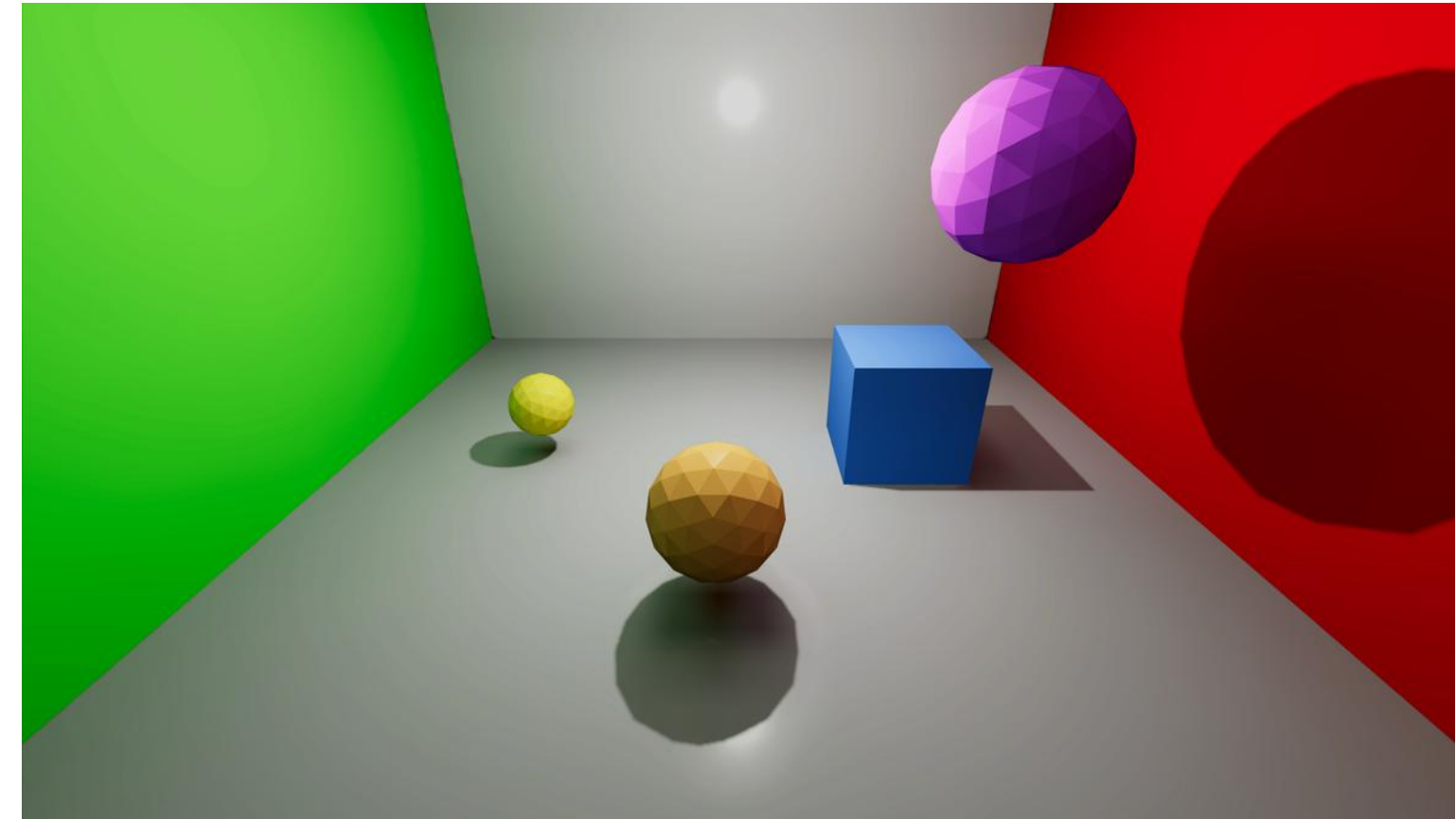
- Computed during the **final scene shading pass**
- Reflection direction computed from the **view vector** and **surface normal**
- **Cone tracing** performed along the **reflection direction**
- **Narrower cone angles** used
- Cone width determined by **material roughness**
- Reflected indirect lighting sampled from voxel radiance
- **Reflection quality** is limited by **voxel resolution**



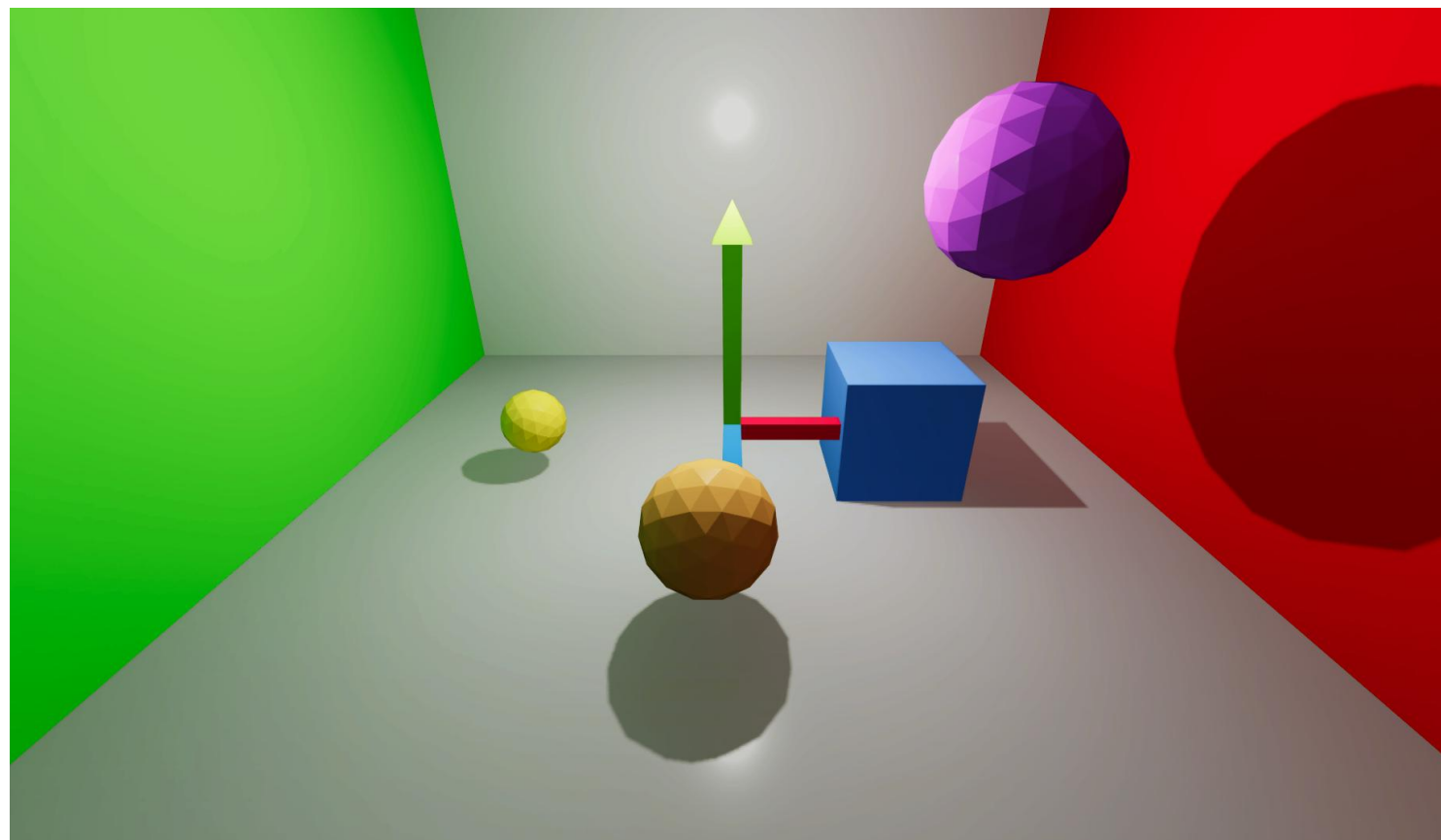
Global Illumination



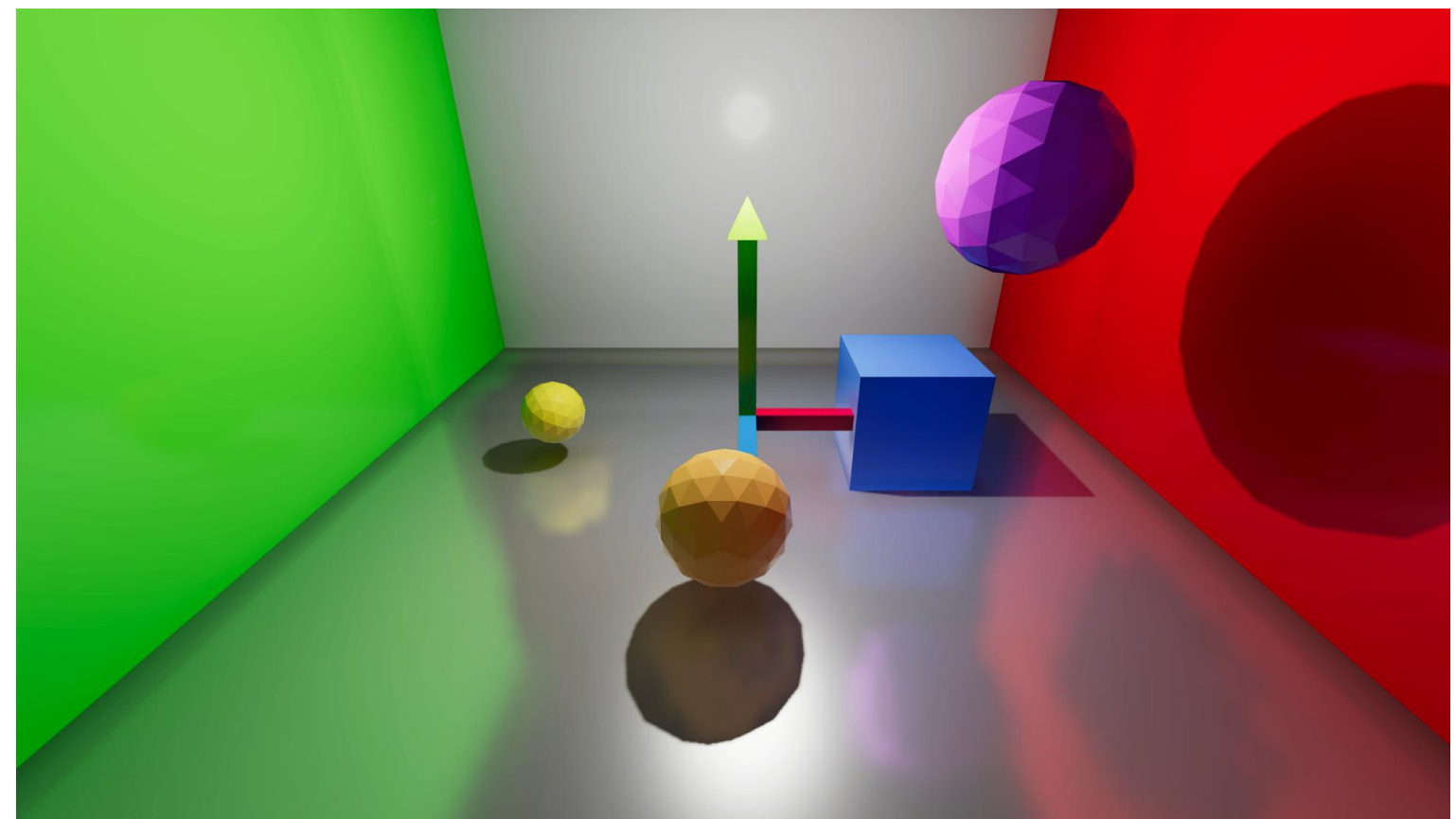
original



DDGI



SurfelGI

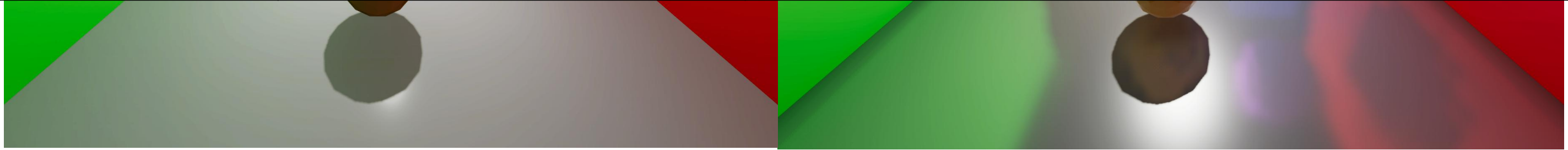


VoxelGI (specular reflection on)

Global Illumination



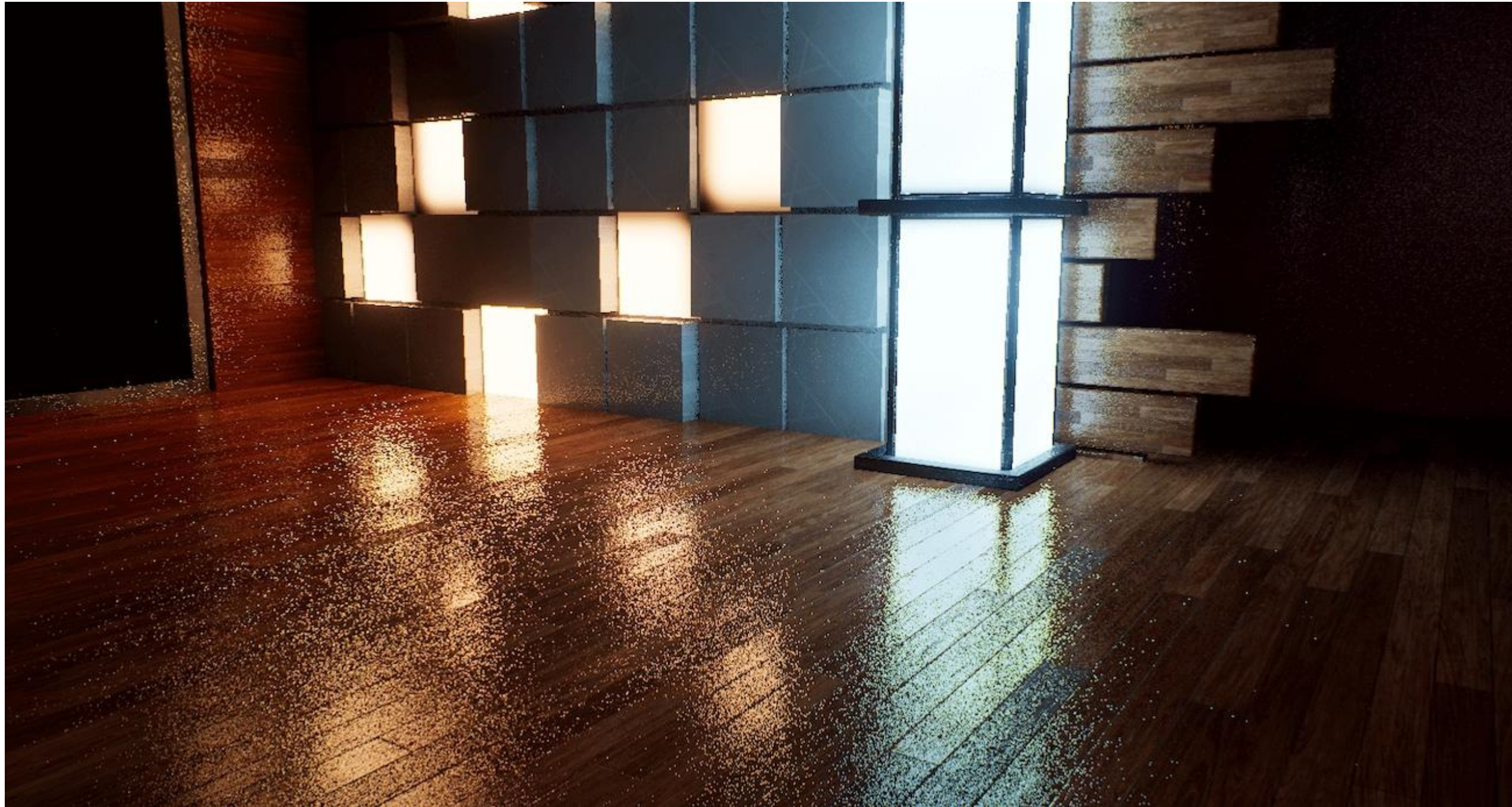
	DDGI	Surfel GI	Voxel GI
Lighting Storage	Probe SH (spatial grid)	Surfel SH (surface-attached)	Voxel Radiance (3D grid)
Indirect Lighting Sampling	8-probe trilinear interpolation	Weighted average of nearby surfels	Cone tracing
Ray Direction	Full 360° directions	Hemisphere around surface normal	Cone with adjustable aperture
Spatial Coverage	Entire world-space probe grid	Only observed surfaces inside the camera frustum	Only voxelized geometry inside the camera frustum
Fine Geometry Handling	Limited by probe density	Limited by surfel density	High detail near the camera (LOD)
Specular Reflection	X	X	O



SurfelGI

VoxelGI (specular reflection on)

Global Illumination



Modern GI techniques often combine multiple GI approaches into a hybrid system

- Example: **Unreal Engine Lumen** combines surface-based lighting caches, distance-field tracing, screen-space reflection, and ray tracing
- Combines strengths of DDGI, Surfel GI, Voxel GI, and other hardware ray tracing techniques